

Exploratory Data Analysis (EDA) Report

Project Overview

This report summarizes the exploratory data analysis performed on the ApexPlanet sales dataset. The objective is to understand the dataset structure, identify data quality issues, analyze distributions, and derive business insights before visualization and statistical analysis.

Dataset Understanding

Key steps:

- Loaded the dataset into a DataFrame.
- Inspected rows, columns and data types.
- Reviewed numerical and categorical attributes.
- Checked dataset dimensions.

Data Cleaning

Verified missing values, duplicate records, inconsistent formats, and converted fields to appropriate data types where required.

Descriptive Statistics

Calculated count, mean, median, minimum, maximum, standard deviation and quartiles for numerical columns to understand the distribution.

Categorical Analysis

Analyzed the frequency distribution of Product, Category, Gender, and City to identify dominant groups.

Sales Analysis

Examined Total Sales across products, categories, and customers. Identified high-performing products and customer segments.

Correlation Analysis

Studied the relationship between numerical variables such as Quantity and Total Sales using correlation and scatter plots.

Visualizations Used

Bar charts, pie charts, line charts, histograms, box plots, scatter plots, and correlation heatmaps were used to reveal patterns.

Key Findings

- Sales are concentrated among a few products/categories.
- Quantity has a positive relationship with Total Sales.
- Some customer segments contribute significantly more revenue.
- Geographic analysis helps identify high-performing cities.

Conclusion

The EDA provides a strong foundation for dashboard creation, hypothesis testing, and business decision-making.

Notebook Notes

Insights

1. Mobile generated the highest revenue.
2. Books was the second-highest selling product.
3. Laptop contributed the least revenue.
4. Revenue is concentrated among a few products.